

A σ -Essential Contextual State: Localizing an Open Problem on Concrete Orthomodular Lattices

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Abstract

A two-valued state on a concrete orthomodular lattice (OML) of subsets of a set Ω assigns a definite truth value to each measurable event, coherently across overlapping measurement contexts. On a Boolean algebra every finitely coherent local pattern of such assignments extends to a global σ -additive two-valued state, and (away from a measurable cardinal) every global one is a point evaluation. We ask whether the non-distributive case can differ: does there exist a concrete σ -complete non-Boolean OML carrying a finitely coherent local pattern that extends to *no* global σ -additive two-valued state — a *σ -essential contextual state*? We show this question *localizes*: a witness exists iff (i) no point evaluation extends the pattern and (ii) no non-principal σ -additive two-valued state does either, and that clause (i) is freely arrangeable (after Navara–Pták), so the entire content sits in clause (ii). We then strip clause (ii) of all lattice scaffolding to a bare set-theoretic existence statement — a non-principal, σ -additive, coherent selection across a system of overlapping countable partitions — and locate it precisely against the surrounding literature: it is empty in the Polish-representable case (Derr–Williamson), it does *not* reduce to the Hilbert-space pure-state dichotomy of Blecher–Weaver (Kochen–Specker; Akemann–Weaver), and it does not reduce to a measurable cardinal in the way its Boolean shadow would. The result is a sharply posed, currently open problem at the interface of quantum logic and the set theory of measure-theoretic structures, together with a map of where its answer must live.

1 Introduction

A finite system of measurement contexts may admit a consistent assignment of truth values to events that no single underlying state realises — the algebraic signature of quantum contextuality. When the contexts are countable and one demands countable additivity, a sharper question arises: can a *finitely* coherent pattern fail to extend to any global σ -additive state, with the failure visible only at the σ -level? This paper isolates that phenomenon for concrete orthomodular lattices, reduces it to an irreducible set-theoretic core, and locates that core against the measure-theoretic and operator-algebraic literature. We work throughout with two-valued states, the non-distributive analogue of a $\{0, 1\}$ -valued measure.

A *concrete σ -complete orthomodular lattice* (concrete σ -OML) is a family $\mathcal{L} \subseteq \mathcal{P}(\Omega)$ with

$$\emptyset, \Omega \in \mathcal{L}; \quad A \in \mathcal{L} \implies A^\perp := \Omega \setminus A \in \mathcal{L}; \quad \{A_n\} \subseteq \mathcal{L} \text{ pairwise disjoint} \implies \bigcup_n A_n \in \mathcal{L},$$

ordered by inclusion. Equivalently it is a σ -class [17], a Gudder concrete quantum logic [10, 18]: the maximal Boolean subalgebras (“blocks”) overlap but their union need not be Boolean. A *two-valued state* is $s: \mathcal{L} \rightarrow \{0, 1\}$ with $s(\Omega) = 1$ that is additive over orthogonal joins lying in its domain: whenever $\{A_n\} \subseteq \mathcal{L}$ are pairwise disjoint with $\bigcup_n A_n \in \mathcal{L}$,

$$s\left(\bigcup_n A_n\right) = \sum_n s(A_n)$$

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(in particular $s(A) + s(A^\perp) = 1$ on complement pairs); it is σ -additive when this holds for countable such families. The same additivity is required of states on a sub-orthoposet $B \subseteq \mathcal{L}$, over orthogonal joins that exist in B . The *point evaluations* $\delta_\omega(A) := \mathbb{1}[\omega \in A]$ are σ -additive two-valued states; call these *Dirac* and all others *non-Dirac*. For a finite $\perp$ -closed sub-orthoposet $B \subseteq \mathcal{L}$ and a two-valued state s_0 on B , s_0 *extends* to s on $\mathcal{L}$ if $s \upharpoonright B = s_0$.

Definition 1.1 (σ -essential contextual state). A σ -essential contextual state on $\mathcal{L}$ is a two-valued state s_0 on a finite $\perp$ -closed sub-orthoposet $B \subseteq \mathcal{L}$ that extends to *no* global σ -additive two-valued state on $\mathcal{L}$. A concrete σ -OML $\mathcal{L}$ *carries a σ -essential contextual state* (is a *witness*) if such an s_0 exists. We require $\mathcal{L}$ to be non-Boolean and *irreducible* (centre $\{\emptyset, \Omega\}$); see Remark 1.2.

The phenomenon is a failure of σ -level compactness. Wright’s theorem [20] — every two-valued state on a *finite* OML is finitely realised — forbids any *finite* witness, so the question is whether a coherent finite pattern can fail to globalise once σ -additivity is imposed. On a Boolean algebra it cannot (§2); we isolate what must differ in the non-distributive case, reduce to the irreducible core, and locate it.

Results. §2: the Boolean baseline (Prop. 2.1, two obstructions). §3: the *Localization Theorem* (Thm. 3.2) — a witness exists iff (i) no Dirac and (ii) no non-Dirac σ -additive two-valued state extends s_0 , with (i) freely arrangeable, localising all weight to (ii), which strips to a bare set-theoretic selection problem (Q. 3.6). §4: the boundary map — empty above the Polish case, irreducible to Blecher–Weaver, irreducible to a measurable cardinal. §5: the open problem, its target sentence, and the constraints localising it.

Remark 1.2 (Irreducibility is not vacuous). The carrier class is inhabited: the horizontal sum MO_κ (κ orthogonal atom-pairs with 0, 1) is concrete, non-Boolean, and irreducible (centre $\{\emptyset, \Omega\}$) for $\kappa \geq 2$ [13], and σ -complete for $\kappa = \omega$. Concreteness does not force a non-trivial centre: by the Zierler–Schlessinger/Gudder criterion it is exactly order-determination by two-valued states [10, 11], unrelated to reducibility. (MO_κ itself is excluded as a witness by Remark 4.2.)

2 The Boolean baseline

On a Boolean σ -algebra, Definition 1.1 is never satisfied. The proof isolates two distinct obstructions a witness must overcome at once; this is the baseline the non-distributive case must escape.

Proposition 2.1 (No witness in the Boolean case). *Let $\mathcal{L} \subseteq \mathcal{P}(\Omega)$ be a Boolean σ -algebra and $B \subseteq \mathcal{L}$ a finite $\perp$ -closed sub-orthoposet with a two-valued state s_0 . Then s_0 extends to a global σ -additive two-valued state on $\mathcal{L}$.*

Proof. Let $B_0 \subseteq \mathcal{L}$ be the finite Boolean subalgebra generated by B . Because $\mathcal{L}$ is Boolean and s_0 is additive over the orthogonal joins in B , s_0 extends to a (finitely additive) two-valued state on B_0 : on a finite Boolean algebra such a state is determined by, and concentrates on, a single atom $a \in B_0$ — the meet of the s_0 -true generators. As a is a nonzero element of the concrete algebra B_0 , it is a nonempty subset of Ω , and

$$K(s_0) := \bigcap \{A \in B : s_0(A) = 1\} \supseteq a \neq \emptyset$$

(we call $K(s_0)$ the *kernel* of s_0). Any $\omega \in K(s_0)$ satisfies $\delta_\omega \upharpoonright B = s_0$ by Lemma 3.1, and δ_ω is σ -additive. So s_0 globalises — and on a concrete Boolean σ -algebra it does so by a Dirac, the case $K(s_0) = \emptyset$ never arising. The extension to B_0 is exactly where distributivity is used; it is unavailable on a non-Boolean carrier, where the s_0 -true generators may have empty meet. $\square$

Two obstructions combine, both of which a witness must defeat:

- (a) (*used above*) compatible events of B generate a finite Boolean algebra whose s_0 -carrying atom is a nonempty common point. A witness breaks this — the events lie in incompatible blocks with empty meet, so no Dirac extends s_0 .
- (b) (*complementary*) away from a measurable cardinal, every global σ -additive two-valued state on a Boolean σ -algebra is Dirac [19, 12]; so even where (a) fails, no *non*-Dirac state repairs the gap on a Boolean carrier.

The two pull against each other: breaking (a) requires non-distributivity, which is precisely what makes (b) escapable by admitting non-Dirac global states. A witness must break (a) while leaving no such repair available.

3 The Localization Theorem

When a Dirac extends s_0 is elementary.

Lemma 3.1. *Let $B \subseteq \mathcal{L}$ be a finite $\perp$ -closed sub-orthoposet with two-valued state s_0 . A point evaluation δ_ω satisfies $\delta_\omega \upharpoonright B = s_0$ iff $\omega \in K(s_0) = \bigcap \{A \in B : s_0(A) = 1\}$. Consequently no Dirac extends s_0 iff $K(s_0) = \emptyset$.*

Proof. $\delta_\omega \upharpoonright B = s_0$ means $\omega \in A \iff s_0(A) = 1$ for every $A \in B$. The forward constraints place $\omega \in \bigcap \{A : s_0(A) = 1\}$. For a false A (i.e. $s_0(A) = 0$), $\perp$ -closure gives $A^\perp \in B$, and additivity gives $s_0(A^\perp) = 1 - s_0(A) = 1$; so $A^\perp$ is one of the true elements and the constraint $\omega \notin A$ is already $\omega \in A^\perp$, subsumed in $K(s_0)$. Thus the false constraints add nothing, and $\delta_\omega \upharpoonright B = s_0 \iff \omega \in K(s_0)$. Conversely any $\omega \in K(s_0)$ works: true A give $\omega \in A$, false A give $\omega \in A^\perp$ hence $\omega \notin A$. $\square$

Theorem 3.2 (Localization). *Let $\mathcal{L}$ be a concrete σ -OML, $B \subseteq \mathcal{L}$ a finite $\perp$ -closed sub-orthoposet, and s_0 a two-valued state on B . Then s_0 is a σ -essential contextual state iff*

- (i) $K(s_0) = \emptyset$ (no point evaluation extends s_0), and
- (ii) no non-Dirac σ -additive two-valued state on $\mathcal{L}$ extends s_0 .

Moreover clause (i) is freely arrangeable: there is a concrete σ -OML and an s_0 with $K(s_0) = \emptyset$.

Proof. The Dirac/non-Dirac dichotomy on global σ -additive two-valued states is exhaustive, so

$$(\text{no extension of } s_0) \iff (\text{no Dirac extends } s_0) \wedge (\text{no non-Dirac extends } s_0).$$

By Lemma 3.1 the first conjunct is (i); the second is (ii). For arrangeability of (i), the Navara–Pták construction (Example 3.4) realises $K(s_0) = \emptyset$ explicitly. $\square$

Remark 3.3 (Scope: a decomposition, not a reduction). The equivalence is a decomposition, not a reduction to a separate named problem: the Dirac/non-Dirac split being exhaustive, it merely unfolds the definition of witness. Its content is that clause (i) is *freely arrangeable*, so all weight of Definition 1.1 sits in clause (ii) — the σ -point-selection core, the subject of the remainder.

The Navara–Pták construction both supplies clause (i) and shows (ii) is binding.

Example 3.4 (Navara–Pták [17]). Let $\Omega = \mathbb{Q}_0 \times \mathbb{Q}_0$ with $\mathbb{Q}_0 = \mathbb{Q} \cap (0, 1)$, and let $f, g: \Omega \rightarrow \mathbb{R}$ be the coordinate projections. Let $\mathcal{L} = \mathcal{L}_{f,g}$ be the least σ -class containing the inverse images of Borel sets under f and g . Set $C_f = f^{-1}(\{1/3\})$, $C_g = g^{-1}(\{1/2\})$, $C_{f+g} = (f+g)^{-1}(\{1/2\})$. Define $m(A) = 1$ iff A contains one of C_f, C_g, C_{f+g} . Navara and Pták show m is a (two-valued, σ -additive) state on $\mathcal{L}$, and that since $C_f \cap C_g \cap C_{f+g} = \emptyset$ the state m is *not* concentrated at any point.

Remark 3.5 (The example is not itself a witness). Let B be the $\perp$ -closure of $\{C_f, C_g, C_{f+g}\}$ and let s_0 be the pattern assigning each of the three the value 1. Since $C_f \cap C_g \cap C_{f+g} = \emptyset$, we have $K(s_0) = \emptyset$, so clause (i) holds — no point evaluation extends s_0 . Yet the Navara–Pták state satisfies $m \upharpoonright B = s_0$ and is non-Dirac and σ -additive, so clause (ii) *fails* and s_0 is not σ -essential: the construction supplies its own rescuer, so clause (i) cannot suffice alone. The repair is available because $\mathcal{L}_{f,g}$ is generated by two *compatible* (commuting) observables — Boolean enough to support a non-Dirac global state through the point-level gap. A witness requires the opposite: a carrier on which no non-Dirac state repairs the gap.

Stripped of lattice scaffolding, this core — clause (ii) — is a selection problem. Each block is a maximal compatible context, equivalently a countable partition Π_α of Ω into elements of $\mathcal{L}$; a two-valued state selects one cell $c_\alpha \in \Pi_\alpha$ per block, subject to coherence, σ -additivity, and non-principality.

Question 3.6 (σ -point-selection, bare form). Given a set Ω and a family $\{\Pi_\alpha\}$ of countable partitions of Ω (the blocks of a concrete σ -OML, pairwise overlapping in their common refinements), does there exist a selection assigning to each Π_α one cell $c_\alpha \in \Pi_\alpha$ such that

- (a) (coherence) the selections agree on events common to several blocks;
- (b) (σ -additivity) the induced $\{0, 1\}$ -valuation commutes with countable disjoint unions;
- (c) (non-principality) there is no $\omega_0 \in \Omega$ with $c_\alpha =$ the cell of Π_α containing ω_0 for all α ;
- (d) (prescription) the selection realises a prescribed finite pattern that no single point realises?

A witness exists iff Question 3.6 has a positive answer for some admissible carrier. Two features of the bare form, obscured in lattice language, fix its position: its distance from a measurable cardinal, and its place within contextuality.

Remark 3.7 (The Boolean shadow is a measurable cardinal; the question is not). For a *compatible* block family (single Boolean base),

$$\begin{aligned} & \text{coherent non-principal } \sigma\text{-additive } \{0, 1\}\text{-selection} \\ &= \text{non-principal countably-complete ultrafilter} = \text{measurable cardinal} \end{aligned}$$

[12]. Question 3.6 imposes coherence (a) across *incompatible, mutually non-distributive* blocks, strictly stronger than an ultrafilter on any single block. The Boolean shadow reduces to a measurable cardinal; the non-distributive coherence is exactly what the reduction cannot cross (§4).

Remark 3.8 (Relation to sheaf-theoretic contextuality). Question 3.6 asserts that locally coherent $\{0, 1\}$ -valuations admit no *global section* — contextuality in the sense of Abramsky–Brandenburger [2], σ -additive form [1]. The framing is standard; what is open is the conjunction the literature does not treat together: countable additivity + a non-Hilbert (abstractly concrete) carrier + a $\{0, 1\}$ -valued selection.

4 The boundary map

We locate Question 3.6 between an upper boundary where it is known empty and the lower routes that fail to settle it.

The upper boundary is the Polish-representable case. Derr and Williamson [6] prove a σ -extension criterion for coherent probabilities on pre-Dynkin systems: when the carrier is *Polish-representable* — Ω Polish, the generated σ -class equal to the Borel σ -algebra, with

inner-regular blocks — finite coherence implies σ -extendability (their Theorem D.6, via Maharam [16]). Specialised to two-valued states, this says that on a Polish-representable carrier every finitely coherent s_0 globalises: *no σ -essential contextual state exists*. A witness, if one exists, must therefore live on a carrier that is *not* Polish-representable.

A Hilbert-space analogue exists, but does not transfer down to the concrete case. On the projection lattice of $B(\ell^2(\kappa))$, Blecher–Weaver [5] show a singular countably additive *pure* state exists iff κ is Ulam measurable — the nearest positive precedent that countable additivity on a non-distributive σ -complete OML *can* carry set-theoretic strength. It does not settle Question 3.6; three obstructions point the same way.

- (1) *Wrong carrier.* For $\dim H \geq 3$ the projection lattice of $B(H)$ admits *no* two-valued state (Kochen–Specker [14]), so it is not concrete; the Blecher–Weaver dichotomy concerns *pure* states, not two-valued ones, which are absent there by Gleason [9].
- (2) *Masa transfer fails.* Carrying a state-existence result to an abelian subalgebra means factoring through a masa; for the pure two-valued case this fails — Akemann–Weaver [3] give a pure state on $B(H)$ multiplicative on no masa, and Blecher–Weaver use Marcus–Spielman–Srivastava paving for exactly this reason.
- (3) *The consumed object is structurally absent.* Blecher–Weaver extract their cardinal from a genuine ultrafilter on the diagonal masa $\ell^\infty(\kappa)$ (projections = subsets, intersection-closed). On a concrete non-Boolean σ -OML, $\mathcal{F}_s := \{A \in \mathcal{L} : s(A) = 1\}$ is an ultrafilter for *no* two-valued state s — not even a filter: meet-closure fails already for δ_ω , since in any concrete MO_2 the atoms a, b have $a \cap b \neq \emptyset$ (else orthogonal) yet $a \wedge b = 0$, so

$$\delta_\omega(a) = \delta_\omega(b) = 1, \quad \delta_\omega(a \wedge b) = 0.$$

The countably-complete ultrafilter an Ulam-style argument would consume is absent from the carrier — the same non-distributivity (lattice meet $\neq$ set intersection), one level down.

Remark 4.1 (Strength of the claim). Remark 3.7 and (1)–(3) are *cited facts assembled*, not a non-reducibility theorem in a fixed reduction calculus. The claim is only that the two obvious bridges — “measurable cardinal” and “transfer from Blecher–Weaver via a masa” — do not close Question 3.6, the obstruction in each being the same non-distributivity. This is what leaves the question open rather than a corollary.

The remaining route from below, faithful representation, is barred by the same obstruction. Routing Question 3.6 through a faithful σ -tribe representation (embedding $\mathcal{L}$ into a σ -algebra of sets, so two-valued states become point evaluations) is unavailable: a faithful tribe-of-sets representation forces distributivity, hence Booleanness (the “Floor” [18]). σ -Loomis–Sikorski holds under the Riesz Decomposition Property (σ -complete MV-algebras, RDP effect algebras [7]), but OMLs lack RDP (MO_2 witnesses), and the available image is only σ -epimorphic, not point-separating. The trivialising route is blocked, again non-distributively.

Remark 4.2 (Segregated carriers do not witness). A carrier is *segregated* when its contextuality is carried by a central Boolean factor — e.g. MO_ω or the product $\prod_n MO_2$, whose σ -additive two-valued states all concentrate at points. On such carriers every finitely coherent s_0 is the restriction of a Dirac, so clause (ii) of Theorem 3.2 fails and there is no witness. A witness must be *non-segregated*: the incompatibility must be irreducible, not routed through a central partition. This is why Definition 1.1 requires irreducibility, and it sharpens the search for a carrier to the non-segregated, non-Boolean, non-Polish-representable regime.

5 The open problem

Assembling §3–§4, a σ -essential contextual state exists iff Question 3.6 has a positive answer on a carrier that is concrete, σ -complete, non-Boolean, irreducible, non-segregated, and non-Polish-representable. The boundary map of §4 settles the answer in no direction: none of the three natural bridges — the Blecher–Weaver dichotomy, a measurable cardinal, or a faithful representation — closes it.

Question 5.1 (Main). Does there exist, in ZFC or under a large-cardinal hypothesis, a concrete σ -complete non-Boolean irreducible OML carrying a σ -essential contextual state? Equivalently, is the existence of a non-principal coherent σ -additive $\{0, 1\}$ -selection across overlapping countable partitions (Question 3.6, non-segregated case) provable, refutable, or independent?

A resolution takes one of three forms, each constrained above:

- *positive* — a carrier in the non-Polish, non-segregated regime; necessarily uncountably generated and non-compositional (Rems. 3.7, 4.1);
- *negative* — extending Derr–Williamson’s Polish emptiness past its topological hypotheses to all concrete σ -OMLs;
- *independence* — two models, with and without the selection; all known precedents (von Neumann–Maharam [4], real-valued measurables [8]) are real-valued and positive, leaving the two-valued non-distributive case open.

To state the target precisely, write $\text{St}(B)$ for the two-valued states on a sub-orthoposet B , $\text{St}_\sigma(\mathcal{L})$ for the global σ -additive two-valued states on $\mathcal{L}$, $\text{Fin}_\perp(\mathcal{L})$ for the finite $\perp$ -closed sub-orthoposets, and Adm for the admissible carriers (concrete, σ -complete, non-Boolean, irreducible). A pattern $s \in \text{St}(B)$ is realised by μ when $\mu \upharpoonright B = s$. The σ -additive *lifting* property and its negation are

$$\begin{aligned}\Phi(\mathcal{L}) &:= \forall B \in \text{Fin}_\perp(\mathcal{L}) \forall s \in \text{St}(B) \exists \mu \in \text{St}_\sigma(\mathcal{L}) \mu \upharpoonright B = s, \\ \neg\Phi(\mathcal{L}) &\equiv \exists B \exists s \in \text{St}(B) \forall \mu \in \text{St}_\sigma(\mathcal{L}) \mu \upharpoonright B \neq s,\end{aligned}$$

the latter asserting a σ -essential contextual state (Def. 1.1, Thm. 3.2). The target sentence is

$$\Psi := \exists \mathcal{L} \in \text{Adm} \neg\Phi(\mathcal{L}) \equiv \exists \mathcal{L} \in \text{Adm} \exists B \exists s \forall \mu \in \text{St}_\sigma(\mathcal{L}) \mu \upharpoonright B \neq s.$$

The leading $\exists \mathcal{L}$ ranges over families of subsets of Ω , not over ω ; by Proposition 2.1 the witness is non-distributive. It is this second-order existential over an unbounded carrier, not the inner arithmetic, that places Ψ beyond any single combinatorial principle.

Remark 5.2 (Constraints on a witness). Each known class of carrier satisfies Φ , so a witness avoids all three:

$$\begin{aligned}\mathcal{L} \text{ Boolean} &\implies \Phi(\mathcal{L}) && (\text{Prop. 2.1}); \\ \mathcal{L} \text{ Polish-representable} &\implies \Phi(\mathcal{L}) && (\text{Derr–Williamson}); \\ \mathcal{L} \text{ segregated (central contextuality)} &\implies \Phi(\mathcal{L}) && (\text{Rem. 4.2}).\end{aligned}$$

Hence a witness is non-Boolean, non-Polish-representable, irreducibly non-central: uncountably generated, non-standard-Borel, genuinely non-distributive. Two facts close the standard methods:

- *Forcing*. A constructible carrier is Polish-representable, so satisfies Φ ; and forcing preserves consistency strength (Lévy–Solovay [15]). It frames independence but cannot produce the object.

- (ii) *σ -Loomis–Sikorski*. The theorem holds only under RDP [7], and a lattice effect algebra has RDP iff it is an MV-algebra; an all-sharp non-Boolean OML lies in the complement, where no representation theorem is known to hold or fail.

Remark 5.3 (Consistency strength of Ψ). On a Boolean base the object of Question 3.6 is a non-principal countably-complete ultrafilter—a measurable cardinal; on $B(\ell^2(\kappa))$ the analogue has exactly Ulam-measurable strength [5]. Neither bound survives for Ψ :

- *Lower bound fails*. “No measurable $\Rightarrow$ every σ -additive two-valued state is Dirac $\Rightarrow \neg\Psi$ ” assumes an extension theorem for σ -algebras, not σ -classes; it is false in ZFC by the non-Dirac Navara–Pták state (Example 3.4). Granting the premise would make the non-Dirac clause vacuous, favouring Ψ .
- *Upper bound unearned*. Ψ governs two-valued selections, outside the real-valued-measurable machinery; the Blecher–Weaver mechanism does not transfer (§4).

The strength of Ψ is open in both directions, not conjecturally measurable.

Remark 5.4 (Localization of the open problem). No candidate field supplies a construction for the constrained carrier:

- non-separable / Maharam measure algebras: Boolean and real-valued;
- descriptive no-section phenomena (non-smooth equivalence relations, failed co-analytic uniformization, turbulence): standard-Borel carriers, real-valued obstructions;
- the Dynkin-system / concrete-logic lineage: finitary, or Polish hence globalising.

The one structural rhyme is Glimm–Effros: non-smoothness of a countable Borel equivalence relation $\Leftrightarrow$ absence of a two-valued section. Transporting it onto a Gudder concrete logic would supply the missing principle, but no such transport appears in the literature. The witness question is therefore neither closed by an impossibility theorem nor reachable by a borrowed construction; its resolution requires a new object in the non-Polish, non-distributive regime.

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